

INTRODUCTION

When it comes to fiber reinforced composite materials, they excel in providing a high specific strength, stiffness, fatigue and corrosion resistance, and design flexibility when compared to conventional metallic materials. Lamination of the matrices gives significantly improved in-plane mechanical properties, but also creates a lack in the out-of-plane strength. This is mainly due to the lack of fibrous reinforcement along the thickness direction of the matrix. This creates a big design limitation for the fiber reinforced composite materials, and has initiated the delamination between layers of these materials. The delamination process can significantly reduce the local stiffness and strength of a material. While there are many ways of delamination can occur, the method that will be focused on is stitching.

Stitching was particularly effective when it came to enhancing the resistance to delamination of the composite laminates. One reason why it was effective was due to the bridging mechanism developed by the stitches across delaminated interfaces, which restrained the opening or sliding displacements of the cracked surfaces. Another benefit of stitching was the improvement of mode I and mode II fracture resistance results, impact damage resistance, and post impact damage tolerance. A big flaw that stitching created was the deterioration of the in-plane strength properties.

When it comes to the effects of stitching on a fiber reinforced composite material, there is a lot of confusion as to whether they improve or deteriorate the mechanical properties and fatigue life of the materials. In some studies [1] [2] [3] [4] it has been shown that the deterioration and negative effects on stitching for delamination. Stitching was found creating serious deteriorations in the in-plane strength properties in these composites. While in some other studies [5] [6] [7] [8] it has been shown that compressive and tensile static strengths can be unchanged or improved by stitching.

To further test the effects of stitching on fiber reinforced composite materials regarding mechanical properties and fatigue life, various fiber dominated and matrix dominated layouts were carried out. Specifically fiber dominated [$\pm 45/0/90$]s and [$\pm 30/90$]s layups, and matrix dominated [$\pm 30_2/-30_2$]s, [$+45_2/-45_2$]s, and [$+60_2/-60_2$]s layups. Uncured prepreg laminates were stitched by an industrial sewing machine that used the modified lock stitch which can be seen below in Figure 1. The specimens were cut from a cured panel with a size of 20 mm wide, 200 mm long, and 1.2 mm thick.

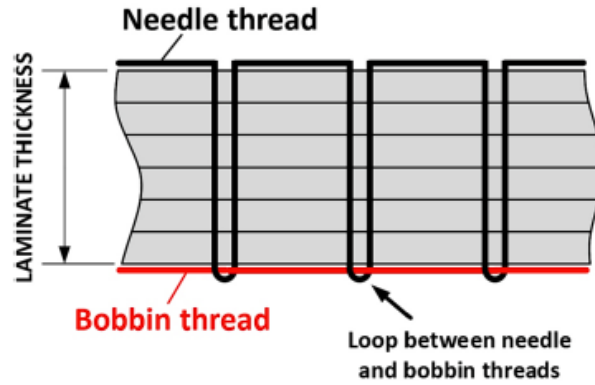


Figure 1: Drawing of lock stitch method used for stitching the various layups.

EXPERIMENTAL RESULTS

For each layup, the strength was obtained by experimenting on at least 3 specimens, while stiffness was measured in the strain range from 0.3×10^{-3} to 3.0×10^{-3} on at least 10 different specimens. The results of this experiment show that the insertion of stitches slightly degrades the stiffness of the fiber-dominated $[0/\pm 45/90]_s$ specimen, while it does not affect the stiffness of matrix-dominated laminates as seen below in Figure 2. Noticeably, there is also a

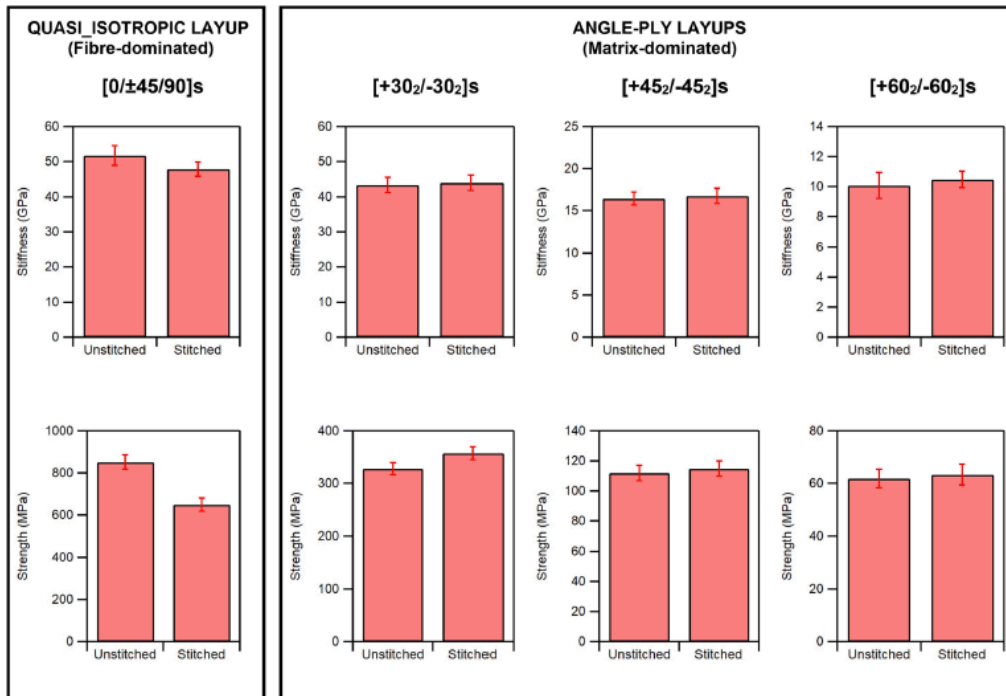


Figure 2: Graphs of stiffness (top row) and strength (bottom row) results after the tests.

large difference in strength when it comes to the stitched vs. unstitched $[0/\pm 45/90]_s$ specimen. Similar to stiffness, there are minimal changes for the other samples, except for very small increases in the stitched results compared to the unstitched. As for the fatigue behavior, as seen below in Figure 3, the different layups of the panels undergo different failure patterns. While the panels have similar fatigue lives (Figure 4), the unstitched samples were subjected to a maximum cyclic stress $\sim 50\%$ higher than that one applied to the stitched sample. Overall they showcase a similar failure pattern, of matrix cracks and delamination initiation at the edges which grow during the fatigue testing.

CONCLUSION

Overall, the results showcase that stitching significantly reduces both the static and fatigue strength of fiber dominated laminates and drastically modifies the damage mechanisms responsible for their final failure. The results also showcase that the effects of stitching can drastically change depending on the angle-ply layups, where in Figure 4 it can be seen that fatigue life is improved for the $[\pm 30_2/-30_2]_s$ and $[+45_2/-45_2]_s$ layups, but for the $[+60_2/-60_2]_s$ the fatigue life is unchanged.

CONNECTION TO LECTURES

This paper greatly emphasizes the angle-ply and stiffness of the matrices. It was shown that while everything may be the same, slight angle differences can greatly affect the performance of the fiber reinforced composite materials. From increasing/decreasing the mechanical properties, to altering the fatigue life, angle-ply should be considered when experimenting with composite materials.

Another connection is for the stitching of the composite materials. This is very similar to the 3-D reinforcements, like braiding and angle-interlock construction. Although the one used here was stitching in a lock stitch configuration, the results were similar, in that the composite had altered material properties than before.

The stress/strain of the matrix is also similar to what we learned in lectures. While not calculated by hand like in class, the deformation and alteration of the composite matrix was looked at and recorded for values to determine the matrix efficiency in different angle-ply. The lamination of the matrices also correlates to what we learned most recently, showcasing the symmetry and evaluation of the matrices and how we define the differences in them, while in every other way they are the same.

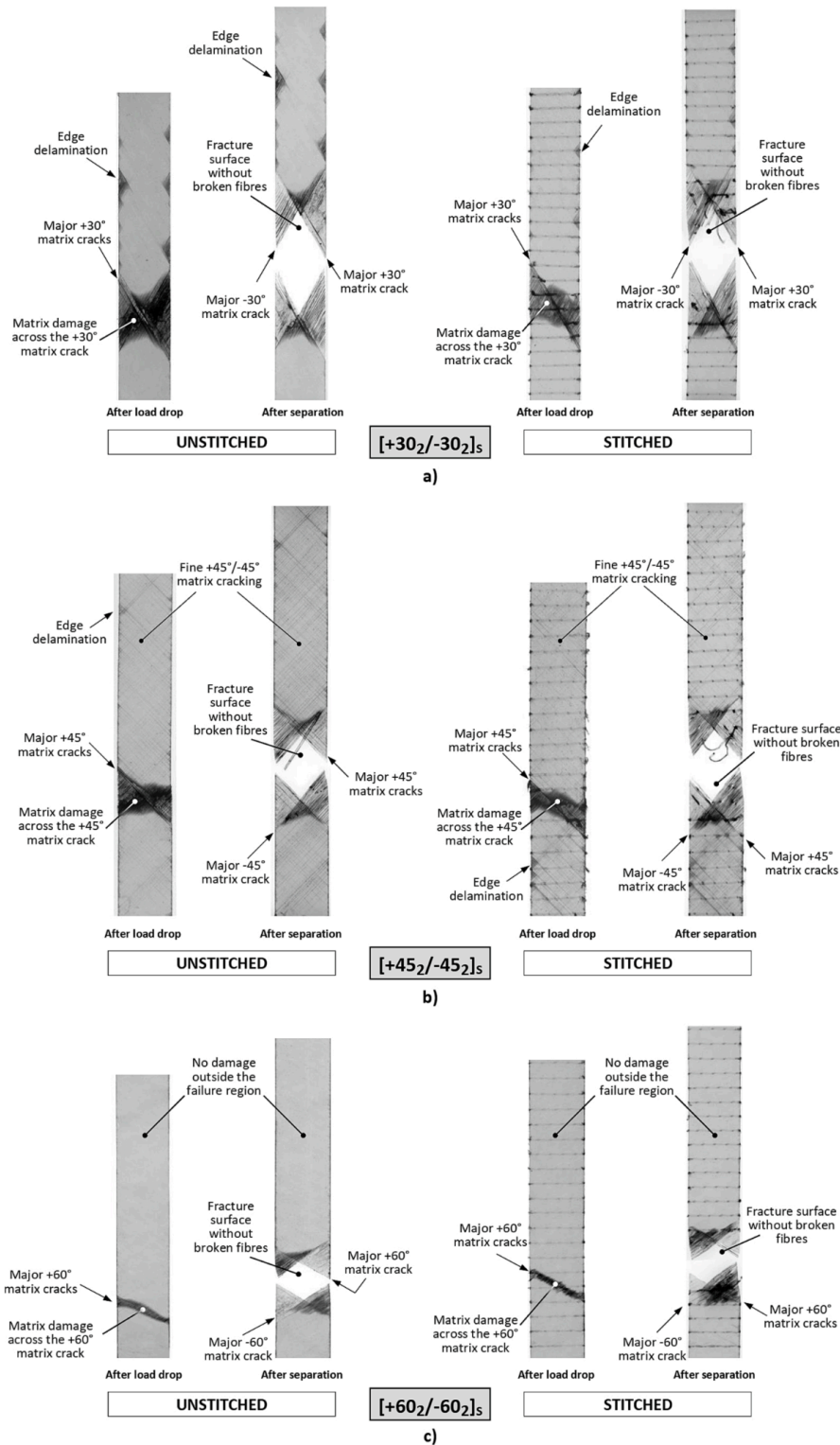
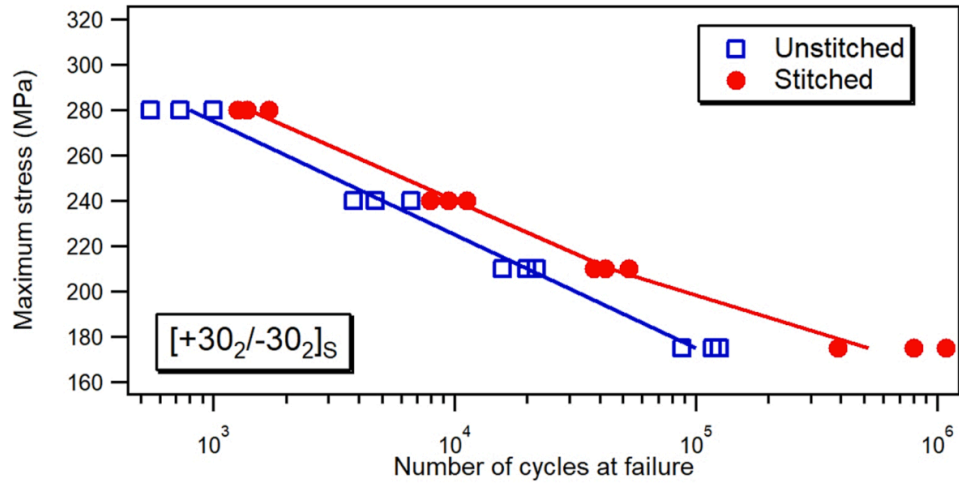
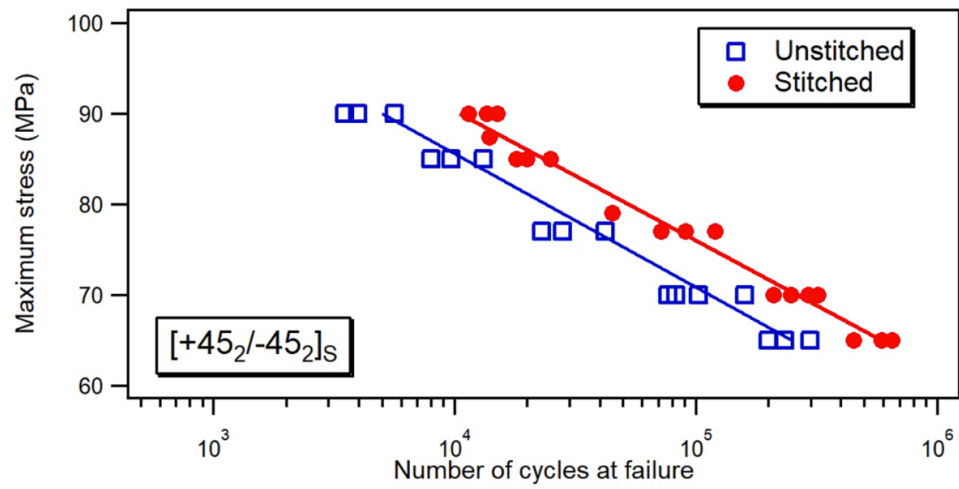


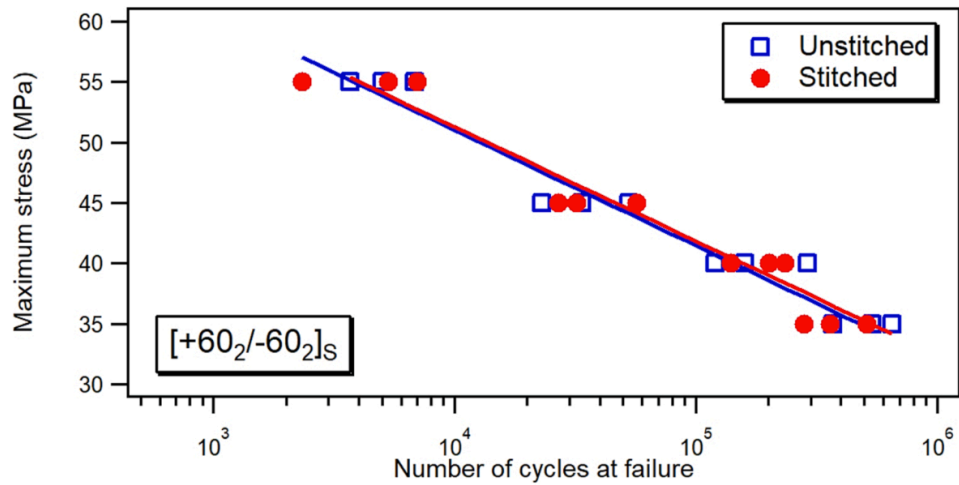
Figure 3: X-radiographs of unstitched and stitched layups and their failure notes.



a)



b)



c)

Figure 4: S-N fatigue data of unstitched and stitched layup samples.

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